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Your grade: 100%

Your latest: 100%

Your highest: 100%

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Learning Check - Linear Regression & Coefficient Uncertainty

Instructions

Review Learning Objectives

1. For the first two questions, we will compute the coefficients of linear regression. A dataset contains the following four points: 1 / 1 point

$D_{xy} = \{(x_i, y_i)\}_{i=1}^4 = \{(0, 1), (0, 3), (10, 10), (10, 6)\}$

coch

Ready to review what you've learned before starting the assignment? I'm here to help.

Because this dataset only contains one predictor ($p = 1$), we will first compute the coefficients using the formula for univariate linear regression.

Let's try

What is β ?

☐ 4

☒ 5

☐ 2

☐ 0.6

Assignment details

Due

Sep 27, 11:59 PM IST

Attempts

4 left (5 attempts every 24 hours)

Submitted

Sep 23, 11:43 PM IST

Try again

☒ Correct

Correct, and this is also $\bar{y}$ as well.

Your grade

To pass you need at least 80%. We keep your highest score.

View submission

See feedback

100%

2. In the same problem, what is β_1 ? 1 / 1 point

☐ 5

☒ 0.6

☐ 2

Like

Dislike

Report an issue

☒ Correct

3. For Questions 3-6, we will find the coefficients using the matrix form expression 1 / 1 point

$\hat{\beta} = (M_X^T M_X)^{-1} M_X^T y$

We define the following:

$\hat{\beta} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix}, M_X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \\ 1 & x_4 \end{bmatrix}, \text{ and } y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$

What is M_X^T ?

☐ $\begin{bmatrix} 0 & 0 & 10 & 10 \end{bmatrix}$

☒ $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 10 & 10 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 10 & 10 & 0 & 0 \end{bmatrix}$

☐ $\begin{bmatrix} 0 & 0 & 10 & 10 \\ 1 & 1 & 1 & 1 \end{bmatrix}$

☒ Correct

4. What is $M_X^T M_X$? 1 / 1 point

☒ $\begin{bmatrix} 4 & 20 \\ 20 & 200 \end{bmatrix}$

☐ $\begin{bmatrix} 20 & 4 \\ 20 & 200 \end{bmatrix}$

☐ $\begin{bmatrix} 20 & 4 \\ 200 & 20 \end{bmatrix}$

☐ $\begin{bmatrix} 4 & 20 \\ 200 & 20 \end{bmatrix}$

☒ Correct

5. What is $(M_X^T M_X)^{(-1)}$? 1 / 1 point

$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Hint: The inverse of a 2×2 matrix is computed as follows:

where $ad - bc$ is called the determinant of the matrix.

☐ $\begin{bmatrix} 1 & 5 \\ 0.1 & 0.25 \end{bmatrix}$

☐ $\begin{bmatrix} 0.1 & -0.05 \\ -0.01 & 0.05 \end{bmatrix}$

☒ $\begin{bmatrix} 0.5 & -0.05 \\ -0.05 & 0.01 \end{bmatrix}$

☐ $\begin{bmatrix} -0.05 & 0.01 \\ 0.5 & -0.05 \end{bmatrix}$

☒ Correct

6. What is $M_X^T y$? 1 / 1 point

☒ $\begin{bmatrix} 20 \\ 100 \end{bmatrix}$

☐ $\begin{bmatrix} 100 \\ 20 \end{bmatrix}$

☐ $\begin{bmatrix} 400 \\ 200 \end{bmatrix}$

☐ $\begin{bmatrix} 20 \\ 200 \end{bmatrix}$

☒ Correct

7. In the following set of equations, what is the interpretation of σ^2 ? 1 / 1 point

$SD(\hat{\beta}_1)^2 = \text{Var}(\hat{\beta}_1 | \{x_i\}_{i=1}^n) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$

$SD(\hat{\beta}_0)^2 = \text{Var}(\hat{\beta}_0 | \{x_i\}_{i=1}^n) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right)$

☐ σ^2 is the standard deviation of the beta coefficients, which follow a normal distribution

☐ σ^2 is the standard deviation of the measurement noise ϵ in a linear additive model

☐ σ^2 is the variance of the beta coefficients arising from matrix multiplication

☒ σ^2 is the variance of the measurement noise ϵ in a linear additive model

☒ Correct

$\sigma^2 = \text{var}(\epsilon)$

https://www.coursera.org/learn/machine-learning-essentials/assignment/submit/091DK/learning-check-linear-regression-coefficient-uncertainty/view-feedback

1/1